November 9, 2026

Jordan A. Awan
jaa557@pitt.edu
Google Scholar

RESEARCH INTERESTS

Data Privacy

Theoretical and applied problems in differential privacy; Statistical inference on privatized data; Theoretical guarantees for synthetic data.

Statistics

Simulation-based inference; Computational statistics, Functional data analysis, Causal inference, U-statistics and kernel tests

Applied Work

Analysis of physiological signals; Acoustic Analyses; Frequency tracking algorithms

EDUCATION

Penn State University, University Park PA **August 2016-May 2020**
Doctor of Philosophy, Statistics.
Advised by Aleksandra Slavković and Matthew Reimherr.

Brandeis University, Waltham MA **Fall 2014-Spring 2016**
Master of Arts, Mathematics.

Clarion University of Pennsylvania, Clarion PA **Fall 2011-Spring 2014**
Bachelor of Science, Mathematics.
Minors: Computer Science, Honors.

PROFESSIONAL CAREER

University of Pittsburgh, Department of Statistics, Pittsburgh PA **August 2025-present**
Assistant Professor.

Purdue University, Department of Statistics, West Lafayette IN **August 2025-present**
Adjunct Professor.

MITRE **June 2021-September 2025, October 2026-present**
Differential privacy consultant for MITRE and Census Disclosure Avoidance System team.

Purdue University, Department of Statistics, West Lafayette IN **August 2020-August 2025**
Assistant Professor.

Harvard University, Center for Research on Computation and Society (CRCS), Cambridge MA **Summer 2018**
Visiting Graduate Student.

Penn State University, Department of Statistics, University Park PA **Summer 2017-Spring 2020**
Research Assistant.
Teaching Assistant (Spring 2019). STAT 401.

Lafayette College, Department of Mathematics, Easton PA **Summer 2013**
REU participant.

HONORS & AWARDS

Journal of Voice 2025 Best Paper Award — **Spring 2026**
Best Paper in the Speech-Language Pathology category

ASHA Convention Visionary Session, American Speech and Hearing Association — **Fall 2024**
Presentation "Estimating Transglottal Airflow Using a Vortex Whistle System and the Phonation Quotient" selected as Visionary

2024 College of Science Recognition Award, Purdue University — **Spring 2024**
Recognized as the recipient of high profile projects

Faculty/Staff Recognition Award, Purdue Statistics — **Spring 2024**

2nd place Best Poster Award, Statistics and Optimization in Data Science Workshop, Purdue University — **Summer 2023**

Outstanding Poster Award, Midwest Machine Learning Symposium — **Spring 2023**

Regina and Norman Carroll Research Award for 2022, Purdue Statistics — **Spring 2023**
Recognized for distinctive contributions to statistical science

Journal of Voice 2022 Best Paper Award — **Spring 2023**
Best Paper in the Speech-Language Pathology category

The Voice Foundation Best Poster Award — **Summer 2021**

Penn State 2020 Alumni Dissertation Award — **Spring 2020**

PSU Statistics 50th Anniversary Best Poster Award — **Spring 2018**

August and Ruth Homeyer Graduate Fellowship, PSU — **Fall 2017-Spring 2018**

Best Performance on Applied Qualifying Exam, PSU Statistics — **Summer 2017**

Stephen B. Brumbach Distinguished Graduate Fellowship, PSU — **Fall 2016-Spring 2017**

GAANN Fellowship — **Fall 2014-Summer 2016**

MAA Outstanding Student Poster Award — **Winter 2014**

Clarion University France-Allison Presentation Award — **Fall 2013**

MAA Outstanding Student Presentation Award — **Summer 2013**

Board of Governors Academic Tuition Scholarship — **Fall 2011-Spring 2014**

SUBMITTED PAPERS & PREPRINTS

"U":undergraduate student, "G": graduate student, and "P": postdoctoral researcher, at the time of the project

1. **Awan, J.**, Barrientos, A. F., Ju, N. "Statistical Inference for Privatized Data with Unknown Sample Size." arXiv:2406.06231.
2. **Awan, J.**, Chen, X., Molinari, R. "Large-Sample Bayesian Approximations for Privatized Data." arXiv:2604.24817.
3. Chen, H., Chen, Y.-W., Song, Q., **Awan, J.** "On the Order of Privacy and Smoothing in Functional Mean Estimation."
4. Chen, Y.-W.^{G}, Pandey, A., **Awan, J.** "Optimal Conversion from Zero-Concentrated to Gaussian Differential Privacy." arXiv:2611.01218.
5. Chen, Y.-W.^{G}, Pasupathy, R., **Awan, J.** "Near-Optimal Private Tests for Simple and MLR Hypotheses." arXiv:2601.21959.
6. Cho, Y.^{G}, **Awan, J.** "Beyond Data Splitting: Full Conformal Prediction by Differential Privacy." arXiv:2603.07522. **2026 ASA Student Paper Award, Statistical Learning and Data Science Section**.

7. Cho, Y.[G], **Awan, J.** "Formal Privacy Guarantees with Invariant Statistics." arXiv:2410.17468.
8. Miglioli, C.[P], **Awan, J.** "Incomplete U-Statistics of Equireplicate Designs: Berry–Esseen Bound and Efficient Construction." arXiv:2510.20755.
9. Moosavi, B.[G], **Awan, J.** "Public Good Provision under Locally Private Signals." arXiv:2606.24013.
10. Ohnishi, Y.[G], **Awan, J.** "Differentially Private Covariate Balancing Causal Inference." arXiv:2410.14789.
11. Tripp, R., Shelly, S., **Awan, J.**, Gillespie, A. "Delivery of Conversation Training Therapy (CTT) in the Group Therapy Environment."
12. Wang, Z.[G], Chang, A.[G], **Awan, J.** "Optimal Debiased Inference on Privatized Data via Indirect Estimation and Parametric Bootstrap." arXiv:2507.10746.

REFEREED PUBLICATIONS

1. Awan, S., **Awan, J.**, McKenna, V., Gifford, S., Burns, E., Sun, Z.[G], Chen, J., Eddins, D. (2026+) "Effect of Vowel Type on Aerodynamic Measures of Estimated Subglottal Pressure and Airflow." *Journal of Speech, Language, and Hearing Research*. Accepted.
2. Awan, S., **Awan, J.**, McKenna, V., Gifford, S., Fletcher Diggs-White, B., Burns, E., Eddins, D., Gillespie, A., Aljowaiser, O., Sun, Z., Chen, J. (2026+) "A Novel High Flow Vortex Whistle System for Measurement of FVC and FEV1." *Journal of Voice*. Accepted.
3. Eng, K.[G], **Awan, J.**, Ju, N., Rao, V., Gong, R. (2026+) "dapper: Data Augmentation for Private Posterior Estimation in R." *R Journal*. Accepted.
4. Chen, Y.-W.[G], Sanghi, P.[U], **Awan, J.** (2026) "Particle Filter for Bayesian Inference on Privatized Data." *Journal of Machine Learning Research*. Accepted.
5. Awan, S., **Awan, J.** (2025+) "Comparison of Methods of Eliciting Vital Capacity: Forced vs. Slow Vital Capacity." *Journal of Voice*. Accepted and available online.
6. Awan, S., **Awan, J.**, Bonilha, H., Gillespie, A., McKenna, V., Chen, J., Eddins, D. (2025+) "Using a Vortex Whistle System to Estimate Phonatory Airflow via the Phonation Quotient." *Journal of Voice*. Accepted and available online.
7. **Awan, J.**, Cai, Z. (2025) "One Step to Efficient Synthetic Data." *Statistica Sinica*. Volume 35, Pages 531-569.
8. **Awan, J.**, Edwards, A., Bartholomew, P., Sillers, A. (2025) "Best Linear Unbiased Estimate from Privatized Histograms." *Journal of Machine Learning Research*. Volume 26, Issue 174, Pages 1-41.
9. **Awan, J.**, Wang, Y.[U] (2025) "Differentially Private Kolmogorov-Smirnov-Type Tests." *Electronic Journal of Statistics*. Volume 19, No. 1, Pages 718-744.
10. **Awan, J.**, Wang, Z.[G] (2025) "Simulation-Based Finite-Sample Inference for Privatized Data." *Journal of the American Statistical Association*. Pages 1-14.
11. Awan, S., Shaikh, M., **Awan, J.,** Abdalla, I., Lim, K., Misono, S., (2025) "Smartphone Recordings are Comparable to 'Gold Standard' Recordings for Acoustic Measurements of Voice." *Journal of Voice*. Volume 39, Issue 4, Pages 1019-1032. **Best Paper Award**.
12. Chen, Y.-W.[G], Pasupathy, R., **Awan, J.** (2025) "Optimal Survey Design for Private Mean Estimation." *Proceedings of the 42nd International Conference on Machine Learning*. Pages 8856–8870.
13. Feinstein, H.[G], Daşdöğen, Ü.[G], **Awan, J.,** Awan, S., Verdolini Abbott, K. (2025) "Comparative Analysis of Two Methods of Perceptual Voice Assessment." *Journal of Voice*. Volume 39, Issue 4, Pages 1065-1075.
14. Ohnishi, Y.[G], **Awan, J.** (2025) "Locally Private Causal Inference for Randomized Experiments." *Journal of Machine Learning Research*. Volume 26, No. 14, Pages 1-40.
15. Wang, Z.[G], Cheng, G., **Awan, J.** (2025) "Differentially Private Bootstrap: New Privacy Analysis and Inference Strategies." *Journal of Machine Learning Research*. Volume 26, Issue 257, Pages 1-57.
16. **Awan, J.**, Bernardi, O. (2024) "Tutte Polynomials for Regular Oriented Matroids." *Discrete Mathematics*. Volume 347, Number 1.
17. **Awan, J.**, Ramasethu, A.[G] (2024) "Optimizing Noise for f-Differential Privacy via Anti-Concentration and Stochastic Dominance." *Journal of Machine Learning Research*. Volume 25, Number 351, Pages 1-32.

18. Kang, T.[G], Kim, S.[G], Sohn, J.[G], **Awan, J.** (2024) "Differentially Private Topological Data Analysis." *Journal of Machine Learning Research*. Volume 25, No. 189, Pages 1-42.
19. **Awan, J.**, Rao, V. (2023) "Privacy-Aware Rejection Sampling." *Journal of Machine Learning Research*. Volume 24, No. 74, Pages 1-32.
20. **Awan, J.**, Vadhan, S. (2023) "Canonical Noise Distributions and Private Hypothesis Tests." *Annals of Statistics*. Volume 51, Number 2, Pages 547-572.
21. **Awan, J.**, Dong, J. (2022) "Log-Concave and Multivariate Canonical Noise Distributions for Differential Privacy." *Advances in Neural Information Processing Systems 36*, 34229-34240.
22. **Awan, J.**, Frechette, C., Li, Y., McMahon, E. (2022) "Demicaps in $AG(4,3)$ and their Relation to Maximal Cap Partitions." *Graphs and Combinatorics*. Volume 83, No. 193.
23. Awan, S., **Awan, J.** (2022) "Use of a Vortex Whistle for Measures of Respiratory Capacity." *Journal of Voice*. Volume 36, Issue 5, Pages 630-636. **Best Paper Award**.
24. Ju, N., **Awan, J.**, Gong, R., Rao, V. (2022) "Data Augmentation MCMC for Bayesian Inference from Privatized Data." *Advances in Neural Information Processing Systems 36*, 12732-12743.
25. Li, A.[G], **Awan, J.**, Chen, J., Eddins, D., Awan, S. (2022) "Enhancing the Vortex Whistle for Measures of Respiratory Capacity via CFD and CAA." *Journal of Biomechanical Engineering*. Volume 144, Issue 11.
26. Li, A.[G], Chen, J., **Awan, J.**, Eddins, D., Awan, S. (2022) "Performance Analysis and Parametric Study of Vortex Whistle." *Proceedings of the ASME 2022 Fluids Engineering Division Summer Meeting. Volume 1: Fluid Applications and Systems (FASTC); Fluid Measurement and Instrumentation (FMITC); Fluid Mechanics (FMTC)*. Toronto, Ontario, Canada. August 3–5, 2022. V001T01A018. ASME.
27. **Awan, J.**, Slavković, A. (2021) "Structure and Sensitivity in Differential Privacy: Comparing K-Norm Mechanisms." *Journal of the American Statistical Association*. Volume 116, Number 534, 935-954.
28. **Awan, J.**, Bernardi, O. (2020) "Tutte Polynomials for Directed Graphs." *Journal of Combinatorial Theory, Series B*. Volume 140, 192-247.
29. **Awan, J.**, Slavković, A. (2020) "Differentially Private Inference for Binomial Data." *Journal of Privacy and Confidentiality*. Volume 10, No. 1.
30. Awan, S., **Awan, J.** (2020) "A Two-Stage Cepstral Analysis Procedure for the Classification of Rough Voices." *Journal of Voice*. Volume 34, Issue 1, 9-19.
31. **Awan, J.**, Kenney, A., Reimherr, M., Slavković A. (2019) "Benefits and Pitfalls of the Exponential Mechanism with Applications to Hilbert Spaces and Functional PCA." *Proceedings of the 36th International Conference on Machine Learning*, 97:374-384.
32. Reimherr, M., **Awan, J.** (2019) "Elliptical Perturbations for Differential Privacy." *Advances in Neural Information Processing Systems 33*. 10185-10196.
33. Reimherr, M., **Awan, J.** (2019) "KNG: The K-Norm Gradient Mechanism." *Advances in Neural Information Processing Systems 33*. 10208-10219.
34. **Awan, J.**, Slavković, A. (2018) "Differentially Private Uniformly Most Powerful Tests for Binomial Data." *Advances in Neural Information Processing Systems 32*, 4208-4218.
35. Gaskill, C., **Awan, J.**, Watts, C., Awan, S. (2017) "Acoustic and Perceptual Classification of Within-sample Normal, Intermittently Dysphonic, and Consistently Dysphonic Voice Types." *Journal of Voice*, Volume 31, Issue 2, Pages 218-228.
36. Awan, S., **Awan, J.** (2013) "The Effect of Gender on Measures of Electroglottographic Contact Quotient." *Journal of Voice*, Volume 27, Issue 4, 433-440.

BOOK CHAPTERS

1. **Awan, J.**, Gong, R. (2024). "Statistical Inference and Differential Privacy." *In* Drechsler, J., Kifer, D., Reiter, J., & Slavković, A. (Eds.), *Handbook of Sharing Confidential Data: Differential Privacy, Secure Multiparty Computation, and Synthetic Data*. Chapman and Hall/CRC.

OTHER PUBLICATIONS

1. **Awan, J.** (2024). "Here's How Machine Learning can Violate your Privacy." *The Conversation.* May 23, 2024.
2. Habib, S., Pires, B., Benedetto, G., Rodriguez, R., **Awan, J.**, Stanley, J., Totty, E., Germinario, G., & Stevenson, R. (2023). "Automated Synthetic Data Validation: Applying Noise Injection for Disclosure Avoidance." *Joint Statistical Meetings (JSM)*, Toronto, Canada.

3. **Awan, J.**, Reimherr, M., Slavković, A. (2020). "Formal Privacy for Modern Nonparametric Statistics." *CHANCE* 33, No. 4. 43-49.
4. Awan, S., **Awan, J.**, Watts, C., S. Gaskill, C. (2018). "Response to Aichinger and Kubin Re: Letter to the Editor "Acoustic and Perceptual Classification of Within-Sample Normal, Intermittently Dysphonic, and Consistently Dysphonic Voice Types"." *Journal of Voice*. Issue 32, No. 3, 383-384.

GRANTS

NIH R01: Deconstructing Voice Therapy: Towards Enhanced Communication Outcomes, Co-I **2025-2030**

Award number: 1R01DC022026-01A1

Co-investigator.

PI: Dr. Amanda Gillespie.

Co-Is: Drs. Kris Tjaden, Eva van Leer, Anthony Law, Adam Klein

$2,821,462 for 5 years.

Directly responsible for $\approx \$300,000$.

NIH R01: Vital Capacity & Airflow Measurement for Voice Evaluation: A Vortex Whistle System, MPI **2023-2028**

Award number: R01 DC020799-01A1

One of 4 MPIs, along with Drs. Shaheen Awan, Jun Chen, and Amanda Gillespie.

Co-Is: Drs. David Eddins and Victoria McKenna

$3,129,418 for 5 years.

Directly responsible for $\approx \$450,000$.

NSF SES: Simulation-Based Inference for Differential Privacy, PI **2022-2027**

Current award number: SES 2610910

Prior award number to Purdue: SES 2150615

Principal investigator

Co-PI: Dr. Roberto Molinari

$450,000.

PENDING PATENTS

261481 — 2085.227.PR3 - Multi-Channel High-Flow Vortex Whistle Spirometer **2026**

Patent application filed

With Shaheen Awan, Jun Chen, David Eddins, and Zheming Sun

256588 — 2085.227.PR - Low Flow Vortex Whistle Airflow Measurement System **2025**

Provisional patent is pending

With Shaheen Awan, Jun Chen, David Eddins, and Zheming Sun

INVITED CONFERENCE PRESENTATIONS

Keystone State Statistics Symposium, University Park, PA **October 2026**

Simulation-Based, Finite-Sample Inference for Privatized Data

Keystone State Statistics Symposium, University Park, PA **October 2026**

Panelist: Early Career Panel

ENAR 2026, Indianapolis, IN **March 2026**

Simulation-Based, Finite-Sample Inference for Privatized Data

Joint Statistical Meetings, Nashville, TN **August 2025**

Best Linear Unbiased Estimate from Privatized Histograms

Joint Statistical Meetings, Portland, OR **August 2024**

Panelist: Evaluating Statistical Disclosure Control Techniques based on the Risk and Utility of Privacy-Protected Data

25th **Annual CERIAS Security Symposium, Purdue University, West Lafayette, IN** **April 2024**

Valid Statistical Inference on Privatized Data

Statistical Learning and Differential Privacy, Bath U.K. (online) **September 2022**

Data Augmentation MCMC for Bayesian Inference from Privatized Data

Workshop on the Analysis of Census Noisy Measurement Files and Differential Privacy, Rutgers University **April 2022**

Posterior Inference on Privatized Data via Data Augmentation MCMC

Computational & Methodological Statistics Meeting, Online **December 2021**

Canonical noise distributions and private hypothesis tests

Virtual Symposium on Data Privacy, ASA Nevada Chapter **September 2021**

Canonical noise distributions and private hypothesis tests

Computational & Methodological Statistics Meeting in Pisa, Italy **December 2018**

Differentially private UMP hypothesis tests for Bernouilli data

Statistical Society of Canada Annual Meeting, McGill University, Montreal Canada **June 2018**

Optimizing finite sample performance under differential privacy

INVITED DEPARTMENTAL and INDUSTRY SEMINARS

Virginia Tech, Department of Statistics, Blacksburg, VA **March 2026**

Beyond Data Splitting: Full Conformal Prediction by Differential Privacy

University of Pittsburgh, Machine Learning Seminar, Department of Mathematics, Pittsburgh, PA **February 2026**

Differentially Private Bootstrap: New Privacy Analysis and Inference Strategies

University of Pittsburgh, Department of Biostatistics, Pittsburgh, PA **January 2026**

Simulation-Based, Finite-Sample Inference for Privatized Data

Capital One, McLean VA (virtual) **January 2026**

Best Linear Unbiased Estimate from Privatized Contingency Tables

National Institute for Research in Digital Science and Technology (Inria), Montpellier, France (virtual) **November 2025**

Differentially Private Bootstrap: New Privacy Analysis and Inference Strategies

University of Pittsburgh, Department of Statistics, Pittsburgh, PA **December 2024**

Simulation-Based, Finite-Sample Inference for Privatized Data

Auburn University, Department of Mathematics and Statistics, Auburn, AL **April 2024**

Simulation-Based, Finite-Sample Inference for Privatized Data

Air Force Institute of Technology, Department of Mathematics and Statistics, Wright-Patterson Air Force Base, OH **January 2023**

Bayesian Inference on Privatized Data

Auburn University, Statistics and Data Science Seminar, Online **September 2022**

Bayesian Inference from Privatized Data

Michigan State University, Department of Statistics, Online **November 2021**
Canonical noise and private hypothesis tests

University of Wisconsin-Madison, Department of Statistics, Madison WI **February 2020**
Differentially private inference for binomial data

Lafayette College, Department of Mathematics, Easton PA **February 2020**
Differentially private inference for binomial data

George Mason University, Department of Statistics, Fairfax VA **February 2020**
Differentially private inference for binomial data

Bucknell University, Department of Mathematics, Lewisburg PA **January 2020**
Differentially private inference for binomial data

Purdue University, Department of Statistics, West Lafayette IN **January 2020**
Differentially private inference for binomial data

Stochastic Modeling and Computational Statistics Seminar at Penn State, University Park PA **February 2018**
Structure and sensitivity in DP: comparing K-korm mechanisms

MIT Combinatorics Seminar, Cambridge MA **April 2016**
Tutte polynomials for directed graphs and oriented matroids

Brandeis Combinatorics Seminar, Waltham MA **January 2016**
Tutte polynomials for directed graphs and oriented matroids

Clarion University Honors Presentations, Clarion PA **Fall 2013**
Results on demicaps in $AG(4, 3)$

CONTRIBUTED CONFERENCE PRESENTATIONS

Joint Statistical Meetings, Toronto Canada **August 2023**
Simulation-Based Inference for Privatized Data

Joint Statistical Meetings, Washington D.C. **August 2022**
Posterior Inference on Privatized Data via Data Augmentation MCMC

Privacy in Machine Learning, Virtual NeurIPS Workshop **December 2021**
Canonical noise distributions and private hypothesis tests

Privacy in Machine Learning, Virtual NeurIPS Workshop **December 2021**
Privacy-aware rejection sampling

Privacy Preserving Machine Learning, Virtual ACM CCS Workshop **November 2021**
Canonical noise and private hypothesis tests

2021 Joint Statistical Meetings, Online **August 2021**
Approximate co-sufficient sampling with applications to goodness of fit tests and synthetic data

2020 Joint Statistical Meetings, Online **August 2020**
KNG: The K-norm gradient mechanism

2019 Joint Statistical Meetings, Denver CO **July 2019**
Benefits and pitfalls of the exponential mechanism

36th International Conference Machine Learning, Long Beach CA **June 2019**

Benefits and pitfalls of the exponential mechanism

Simons Institute for the Theory of Computing, Berkeley CA **April 2019**
Differentially private UMP hypothesis tests for Bernouilli data

2018 Joint Statistical Meetings, Vancouver Canada **July 2018**
Optimizing finite sample performance under differential privacy

Mathematical Foundations of Data Privacy, Banff International Research Station (BIRS), Banff Canada **May 2018**
Structure and sensitivity in DP: comparing K-norm mechanisms

Pi Mu Epsilon Conference, Youngstown OH **Spring 2013**
Maximal caps and substructures in $AG(3,3)$

Joint Math Meetings, Baltimore MD **Winter 2013**
Maximal caps and substructures in $AG(3,3)$

Mathfest Conference, Hartford CT **Summer 2013**
Maximal caps and substructures in $AG(4,3)$

POSTERS

Thirty-Sixth Conference on Neural Information Processing Systems, New Orleans, LA (online) **November 2022**
Log-Concave and Multivariate Canonical Noise Distributions for Differential Privacy

Thirty-Sixth Conference on Neural Information Processing Systems, New Orleans, LA (online) **November 2022**
Data Augmentation MCMC for Bayesian Inference from Privatized Data

Privacy in Machine Learning, Virtual NeurIPS Workshop **December 2021**
Canonical noise distributions and private hypothesis tests

Privacy in Machine Learning, Virtual NeurIPS Workshop **December 2021**
Privacy-aware rejection sampling

Privacy Preserving Machine Learning, Virtual ACM CCS Workshop **November 2021**
Canonical noise and private hypothesis tests

Privacy Preserving Machine Learning, Virtual ACM CCS Workshop **November 2021**
Privacy-aware rejection sampling

Thirty-Third Conference on Neural Information Processing Systems, Vancouver Canada **December 2019**
Elliptical perturbations for differential privacy

Thirty-Third Conference on Neural Information Processing Systems, Vancouver Canada **December 2019**
K-Norm gradient mechanism for private empirical risk minimization

36th International Conference on Machine Learning, Long Beach CA **Summer 2019**
Benefits and pitfalls of the exponential mechanism

Thirty-second Conference on Neural Information Processing Systems, Montreal Canada **December 2018**
Differentially private uniformly most powerful tests for binomial data

Theory and Practice of Differential Privacy in 25th ACM Conference on Computer and Communications Security, Toronto Canada **October 2018**

Differentially private uniformly most powerful tests for binomial data

50th Anniversary Conference at Penn State Department of Statistics, University Park PA **May 2018**

Optimizing finite sample performance under differential privacy

Rao Prize Conference at Penn State, University Park PA **May 2017**

Maximum likelihood estimation with differential privacy

Joint Math Meetings, Baltimore MD **Winter 2014**

REU results on maximal caps and substructures in $AG(4,3)$

OTHER PRESENTATIONS

Lilly Purdue Statistics Seminar, Eli Lilly and Company, Indianapolis IN **Spring 2024**

Statistical Inference with Differential Privacy

Open DP Community Workshop **Summer 2020**

Lightning talk on Binomial inference under differential privacy

Penn State Statistics Graduate Student Association Workshop **Fall 2018**

Introduction to differential privacy

Center for Research on Computation and Society, Harvard University **Summer 2018**

Introduction to differential privacy

Penn State Statistics Graduate Student Association Workshop **Fall 2017**

Introduction to differential privacy

Penn State DS 300: Privacy and Security for Data Sciences **Fall 2017**

Introduction to differential privacy

Brandeis Mathematics Graduate Student Seminar **Fall 2015**

A proof of the 5 color theorem

Brandeis Combinatorics Seminar **Spring 2015**

Topics in matroid representability

Brandeis Mathematics Graduate Student Seminar **Spring 2015**

Topics regarding the Tutte polynomial

Pi Mu Epsilon Conference, Youngstown OH **Spring 2013**

A solution for the 2013 COMAP MCM problem A

Clarion University High School Mathematics Competition **Fall 2012**

Mental math algorithms with proofs and examples

Cumberland Valley Math Modeling Challenge at Shippensburg University **Fall 2011**

A model to predict the economic impacts of different voting systems

TEACHING EXPERIENCE

University of Pittsburgh, Department of Statistics, Instructor **Fall 2025-present**

STAT 3692: Differential Privacy, Fall 2025

STAT 1261/2260: Introduction to Data Science, Spring 2026

Purdue University, Department of Statistics, Instructor **Fall 2020-Spring 2025**

CS/STAT 242: Introduction to Data Science, Spring 2024

STAT 598: Differential Privacy, Fall 2022, Spring 2025

MA/STAT 519: Probability Theory, Fall 2021, Spring 2023, Spring 2025

STAT 692: Research Seminar, Fall 2021, Spring 2022

STAT 417: Statistical Theory, Fall 2020 (2 sections online), Fall 2022

Pennsylvania State University, Department of Statistics, Instructor **Spring 2019**

Introduction to Probability and Statistics with R for Engineers

Brandeis University, Department of Mathematics, Instructor **Fall 2015, Spring 2016**

Integral Calculus

Brandeis University, Department of Mathematics, Grader **Fall 2014, Spring 2015**

Multivariate Calculus, Linear Algebra

Brandeis University, Department of Mathematics, Tutor **Fall 2014, Spring 2015**

Pre-Calculus, Calculus I & II

Clarion University, Department of Academic Enrichment, Tutor **Fall 2011-Spring 2014**

Finite Mathematics, Pre-Calculus, Calculus I & II, Linear Algebra

SOFTWARE DEVELOPMENT

Canonical Noise Mechanism in OpenDP **2022-2024**

Implementation of the canonical noise distributions in "Canonical Noise Distributions and Private Hypothesis Tests" within the OpenDP framework. In collaboration with Aishwarya Ramasethu, Yu-Ju Ku, and Michael Shoemate.

`SimBaRepro`**: Simulation-Based Finite Sample Inference via Repro Samples** **Summer 2025**

R package implementation of the method in "Simulation-Based Finite Sample Inference for Privatized Data." In collaboration with Xinlong Du and Zhanyu Wang. Available on CRAN.

`dapper`**: Data Augmentation for Private Posterior Estimation in R** **Summer 2024**

R package implementation of the method in "Data Augmentation MCMC for Bayesian Inference from Privatized Data." In collaboration with Kevin Eng and Drs. Ruobin Gong, Nianqiao Ju, and Vinayak Rao. Available on CRAN.

`binomialDP`**: Differentially Private Inference for Binomial Data** **Summer 2020**

R package implementation of UMP tests and UMA confidence intervals for Binomial test statistics under differential privacy. In collaboration with Tran Tran and Dr. Aleksandra Slavković.

SERVICE

Journal Referee

Annals of Statistics, Journal of the Royal Statistical Society Series B, Journal of the American Statistical Association, Neural Information Processing Systems, International Conference on Machine Learning, Journal of Privacy and Confidentiality, Journal of Computational and Graphical Statistics, Statistica Sinica, among others

Associate Editor, Journal of Statistical Planning and Inference **Fall 2026-present**

Session Organizer, CFE-CMStatistics 2026 **Spring 2026- present**

Organized a session on statistical inference on privatized data

Poster Judge, Pittsburgh Chapter of American Statistical Association **Spring 2026**

Rated posters by graduate students

Co-Guest-Editor, Statistical Theory and Related Methods **Fall 2025-present**

Special issue on differential privacy

Colloquium Chair, Pitt Statistics **Fall 2025-Spring 2026**

Organized the University of Pittsburgh Statistics seminar, invited speakers

Program Committee Member, Theory and Practice of Differential Privacy **2021, 2024, 2026**
Reviewed submissions and helped organize the workshop

Pre-Examiner for PhD Thesis, University of Helsinki **Summer 2025**
Reviewed the PhD thesis of Ossi Räisä

Session Organizer, Bayesian, Fiducial, & Frequentist Conference (BFF9) **Spring 2025**
Organized a session on statistical inference on privatized data

PhD Qualifying Exam Committee, Purdue Statistics **Spring 2025**
Wrote exam questions and helped to administer the exam

Faculty Mentor, Science Scholars Program, Purdue **Fall 2024-Spring 2025**
Mentored an undergraduate student from an under-represented minority background

Program Committee, AAAI **Fall 2024**
Reviewed articles for the conference

Program Committee, Fairness, Accountability, and Transparency (FAccT) **Spring 2023**
Reviewed submissions for the workshop

Pre-Examiner for PhD Thesis, Aalto University **Spring 2023**
Reviewed PhD thesis of Joonas Jälkö

Colloquium Chair, Purdue Statistics **Fall 2021-Spring 2022**
Organized the Purdue Department of Statistics seminar, invited speakers

Diversity and Inclusion Committee, Purdue Statistics **2021-2025**

Program Committee, NeurIPS Workshop: Privacy and Machine Learning **Fall 2021**
Reviewed submissions and helped organize the workshop

Program Committee, CCS Workshop: Privacy Preserving Machine Learning **Fall 2021**
Reviewed submissions and helped organize the workshop

Graduate Student Admissions, Purdue Statistics **2021-2025**

Distinguished Theme Seminar Series, Purdue Statistics **Fall 2020-Fall 2023**
Member of the organizing committee (Spring 2021-Fall 2021)
Seminar Moderator (Fall 2020, Fall 2021)

Hiring Committee, Purdue Statistics **Fall 2020 - present**
Assistant Professor Search; Escort for interviewees (Fall 2020-Spring 2021)
Assistant and Associate Professor Search (Fall 2021-Spring 2022)
Dream Hire Search (Fall 2023)
Assistant Professor Search (Fall 2024-Spring 2025)

POSTDOCTORAL ADVISOR

Cesare Miglioli, Postdoctoral Researcher **August 2024-August 2026**
Next job: Postdoctoral Researcher in Statistics for Psychology, Section of Psychology, Faculty of Psychology and Educational Sciences (FPSE), University of Geneva

THESIS ADVISOR

Behrooz Moosavi, PhD Student in Economics, Pitt **Summer 2026-present**

Co-advised by Arie Beresteanu

Yu Wei Chen, PhD Student in Statistics, Purdue **Spring 2024-present**

Co-advised by Raghu Pasupathy

Young Hyun Cho, PhD Student in Statistics, Purdue **Spring 2023-Spring 2026**

Co-advised by Will Wei Sun

Note that due to an administrative technicality, I am not listed as co-chair on the thesis

First and current job: Postdoctoral Researcher, Department of Statistics, Harvard University

Yuki Ohnishi, PhD Student in Statistics, Purdue **Summer 2022-Spring 2023**

Co-advised by Arman Sabbaghi

First job: Postdoctoral Associate, Department of Biostatistics, Yale School of Public Health

Current job: Assistant Professor, Department of Statistics, University of Florida

Zhanyu Wang, PhD Student in Statistics, Purdue **Fall 2021-Fall 2023**

Co-advised by Guang Cheng

First and current job: Research Scientist, Meta

OTHER SUPERVISED STUDENTS

Malhar Dave, M.S. Student, Pitt **Summer 2026-present**

Private inference via monotone functions

Feier Chang, PhD Student in Biostatistics, Pitt **Spring 2026-present**

Differentially private and federated learning for generalized linear models

Haotian Chen, PhD Student, Purdue **Fall 2025-present**

Functional data analysis and Gaussian processes for differential privacy

Arin Chang, PhD Student, Purdue **Fall 2024-present**

Indirect inference and parametric bootstrap for privatized data

Hamiltonian MCMC for Bayesian inference on privatized data

Pranav Sanghi, Undergraduate Student, Purdue **Spring 2024-present**

Simulations for Bayesian inference on privatized data

Leo Navarro, Undergraduate Student, Purdue **Summer 2024-Spring 2025**

Optimize Bayesian inference on privatized data

Kefan Gu, Undergraduate Student, Purdue **Summer 2024-Fall 2024**

Optimize Bayesian inference on privatized data

Xinlong Du, M.S. Student, Purdue **Spring 2024-Summer 2025**

R Package development for simulation-based inference

Samuel Forfang, Undergraduate Student, Purdue **Spring 2024**

R Package development for simulation-based inference

Aidan Davis, Undergraduate Student, Purdue **Spring 2024**

R Package development for simulation-based inference

Andrew Liu, M.S. Student, Purdue **Fall 2023-Spring 2025**

Optimize the subsample and aggregate method for confidence intervals

Aishwarya Ramasethu, M.S. Student, Purdue **Fall 2022-Spring 2023**

Research discrete canonical noise distributions and implement binomialDP in OpenDP

Yu-Ju Ku, M.S. Student, Purdue **Summer 2022-Spring 2023**
Implement binomialDP in OpenDP

Burla Ondes, PhD Student in I.E., Purdue **Summer 2022**
Investigated the EM algorithm to analyze privatized data

Taegyu Kang, PhD Student, Purdue **Spring 2022-Summer 2024**
Differentially private topological data analysis (group project)

Sehwan Kim, PhD Student, Purdue **Spring 2022-Summer 2024**
Differentially private topological data analysis (group project)
Formalizing semi-privacy (group project)

Jinwon Sohn, PhD Student, Purdue **Spring 2022-Summer 2024**
Differentially private topological data analysis (group project)

Yue Wang, Undergraduate Student, Purdue **Fall 2021-Summer 2022**
Simulation study to compare differentially private hypothesis tests

Vishnu Suresh (co-chair), M.S. Student, Purdue **Spring 2021-Summer 2021**
Exploring research topics in differential privacy

Jacob Moore, Undergraduate Student, Purdue **Spring 2021-Summer 2021**
Developing an R package for approximate conditional sampling

PHD COMMITTEE

Arin Chang, PhD Student (Statistics, Purdue) **Fall 2025-present**
Advised by Vinayak Rao

Haotian Chen, PhD Student (Statistics, Purdue) **Fall 2025-present**
Advised by Qifan Song

Shubha Sankar Banerjee, PhD Student (Statistics, Pitt)
Advised by Zhao Ren **Summer 2025-present**

Keon Wong Hur, PhD Student (Mechanical Engineering, Purdue) **Spring 2025-present**
Advised by Jun Chen

Hyunwoo Chung, PhD Student (Statistics, Purdue) **Spring 2024-present**
Advised by Fei Xue

Qian Zhang, PhD Student (Statistics, Purdue) **Fall 2022-present**
Advised by Faming Liang

Rajdeep Haldar, PhD Student (Statistics, Purdue) **Spring 2022-Spring 2026**
Advised by Qifan Song

Yi Chu, PhD Student (Statistics, Purdue) **Summer 2022-Summer 2025**
Advised by Raghu Pasupathy

Jiajun Liang, PhD Student (Statistics, Purdue) **Spring 2022-Fall 2023**
Advised by Qifan Song

Xinyi Pei, PhD Student (Statistics, Purdue) **Spring 2021-Spring 2025**
Advised by Vinayak Rao

MS COMMITTEE

Malhar Dave, M.S., Pitt (chair) **Spring 2026-present**
These project on differential privacy

Zheming Sun, M.S. (Mechanical Engineering), Purdue **Fall 2024-present**

Andrew Liu, M.S., Purdue (chair) **Fall 2023-Spring 2025**
Chair of the MS advisory committee. Reading course in differential privacy

Ian Hunter, M.S., Purdue **Spring 2023-Spring 2024**

Nicholas Rosenorn, M.S. (CS & Statistics), Purdue (chair) **Spring 2023-Fall 2024**

Madison Dunn, M.S., Purdue (chair) **Fall 2022-Spring 2024**

Aishwarya Ramasethu, M.S., Purdue **Fall 2022-Spring 2024**
Chair of the MS advisory committee. Reading course in differential privacy

Yu-Ju Ku, M.S. (CS & Statistics), Purdue (chair) **Summer 2022-Spring 2023**

Quisi Zhang, M.S., Purdue **Spring 2022-Spring 2023**

Qi Zhong, M.S., Purdue **Spring 2022-Fall 2022**

Pratiksha Agrawal, M.S., Purdue **Spring 2022-Spring 2023**

Yi-Min Yang, M.S., Purdue **Fall 2021-Spring 2023**

Vidhi Jain, M.S., Purdue **Fall 2021-Fall 2022**

Yu-Wen Wang, M.S., Purdue **Fall 2021-Spring 2023**

Yi-Ting Hung, M.S., Purdue (co-chair) **Fall 2021-Spring 2023**

John Lambrecht, M.S., Purdue (chair) **Spring 2021-Spring 2022**
Chair of the MS advisory committee. Reading course in differential privacy

Vishnu Suresh, M.S., Purdue **Spring 2021-December 2022**

PROFESSIONAL ORGANIZATIONS

Pitt Cyber: Institute for Cyber Law, Policy, and Security, Pitt
Affiliate Scholar **Spring 2026-present**

Privacy and Confidentiality Interest Group, American Statistical Association **Fall 2023-present**
Member

American Statistical Association **Summer 2017-present**

Institute of Mathematical Statistics **Summer 2017-present**

Center for Education and Research in Information Assurance and Security (CERIAS), Purdue **Spring 2024-Summer 2025**
Affiliate faculty member

Regenstrief Center for Healthcare Engineering, Purdue **Summer 2022-Summer 2025**
Associate member